

ABC INSTITUTE OF TECHNOLOGY
AUTONOMOUS EXAMINATIONS BRANCH, HYDERABAD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
B.TECH III YEAR II SEMESTER REGULAR EXAMINATIONS

Branch / Dept: Computer Science and Engineering	Course Code: R20-CSE	Subject Code: CS601PC
Subject: Design and Analysis of Algorithms	Date of Exam: 15-11-2026	Max Marks: 100 Marks
Semester / Year: III Year II Semester	Academic Year: 2026-2027	Time / Duration: 3 Hours

General Instructions:

- 1. Answer all questions in Part A.
- 2. In Part B, answer either (a) or (b) from each question.
- 3. Assume suitable missing data if necessary.

PART A (SHORT ANSWER & CONCEPTUAL BITS)

COMPULSORY BITS

Instructions: Answer all 10 questions. [10 Questions × 2M | Attempt any 10 | Max Marks: 20]

1. Define Introduction: Algorithm and state its significance in Design and Analysis of Algorithms. [2 Marks]
2. Define Disjoint Sets: Disjoint set operations and state its significance in Design and Analysis of Algorithms. [2 Marks]
3. Define Dynamic Programming: General method and state its significance in Design and Analysis of Algorithms. [2 Marks]
4. Define Greedy method: General method and state its significance in Design and Analysis of Algorithms. [2 Marks]
5. Define Branch and Bound: General method and state its significance in Design and Analysis of Algorithms. [2 Marks]
6. What is the primary function or objective of Introduction: Algorithm? [2 Marks]
7. What is the primary function or objective of Disjoint Sets: Disjoint set operations? [2 Marks]
8. What is the primary function or objective of Dynamic Programming: General method? [2 Marks]
9. What is the primary function or objective of Greedy method: General method? [2 Marks]
10. What is the primary function or objective of Branch and Bound: General method? [2 Marks]

PART B (DESCRIPTIVE THEORY & PROBLEMS)

UNIT-WISE ANALYTICAL QUESTIONS (WITH INTERNAL CHOICE)

Instructions: Answer either (a) or (b) from each question. [5 Questions × 16M | Attempt any 5 | Max Marks: 80]

11. (a) How is Introduction: Algorithm implemented in modern software systems? Explain with pseudocode or schema. [16 Marks]

[OR]

- (b) Provide a comprehensive, in-depth analysis of Introduction: Algorithm. Discuss its algorithmic or theoretical design in detail. [16 Marks]

12. (a) Illustrate the step-by-step mechanism of Disjoint Sets: Disjoint set operations using a concrete example. **[16 Marks]**

[OR]

(b) Discuss the trade-offs in time, space, and scalability when engineering systems centered around Disjoint Sets: Disjoint set operations. **[16 Marks]**

13. (a) Explain the fundamental working principle of Dynamic Programming: General method with suitable illustrations. **[16 Marks]**

[OR]

(b) Discuss the trade-offs in time, space, and scalability when engineering systems centered around Dynamic Programming: General method. **[16 Marks]**

14. (a) Explain the mathematical or logical formulation underlying Greedy method: General method. **[16 Marks]**

[OR]

(b) Discuss the trade-offs in time, space, and scalability when engineering systems centered around Greedy method: General method. **[16 Marks]**

15. (a) Explain the mathematical or logical formulation underlying Branch and Bound: General method. **[16 Marks]**

[OR]

(b) Critically evaluate Branch and Bound: General method against alternative paradigms. When is Branch and Bound: General method strictly optimal? **[16 Marks]**